

```
theory Week_1_sol
imports Main
begin
```

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Term 1

Computer-Assisted Theorem Proving: 7CCSACTP

Exercise Sheet 1

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called Week_1.thy and add the following to its beginning.

```
theory Week_1
imports Main
begin
```

Exercise 1.1 Trying out Isabelle

In this sheet, you should try out the Isabelle theorem prover. Prove the following theorem, first on pen-and-paper, then formally in Isabelle.

lemma

```
assumes T: T b
and A: A a  $\wedge$  A b  $\implies$  a = b
and TA:  $\bigwedge x. T x \implies A x$ 
and Aab: A a
shows T a
```

proof—

— *proof* — starts a structured proof without applying any method, leaving the goal unchanged.
— Style: declarative (Isar *have/show*) with apply-style steps; forward (*OF*); backward (*rule*); *subst*.

have 1: A b

— *have* states an intermediate fact (here named 1) and proves it.

using $TA[OF\ T]$
 — *using* passes facts to the proof method that follows.
 — $r[OF\ f]$ discharges the first premise of rule r with fact f (forward reasoning); here $TA[OF\ T]$ is $A\ b$.
 .
 — . proves the goal directly by a fact given with *using*, instantiating its variables if needed.

have $2: A\ a \wedge A\ b$
apply(*rule conjI*)
 — *rule r* matches the conclusion of r with the goal and replaces the goal by r 's premises (backward reasoning); here *conjI* leaves $A\ a$ and $A\ b$.
using $Aab\ 1$
 .

have $3: a = b$
apply(*rule A*)
using 2
 .

show *?thesis*
 — *show* proves the goal of the enclosing proof; *?thesis* abbreviates that goal.
using T
apply(*subst 3*)
 — *subst eq* rewrites the goal with the equation *eq*, left to right; here a becomes b .
 .
qed
 — *qed* closes the proof block, once *show* has proved its goal.

end